

Fractional indices

A Warm up! Simplify:

1 $m^4 \times m^5$

2 $3a^5 \times 2a^3$

3 $y^{10} \div y^5$

4 $8x^7 \div 4x^3$

5 $x^4y^6 \div xy^3$

6 $(a^2)^4$

7 $(3y^2)^3$

8 $m^2 \div m^6$

9 $y^4 \times y^{-2}$

10 $12x^{-3} \div 2x^{-1}$

Fractional Indices mean 'taking the root of'

$$a^{\frac{1}{n}} = \sqrt[n]{a} \quad \text{eg } m^{\frac{1}{3}} = \sqrt[3]{m}$$

B Find the value of:

1 $25^{\frac{1}{2}}$

2 $49^{\frac{1}{2}}$

3 $8^{\frac{1}{3}}$

4 $125^{\frac{1}{3}}$

5 $16^{\frac{1}{4}}$

6 $27^{\frac{1}{3}}$

7 $64^{\frac{1}{2}}$

8 $16^{\frac{1}{4}}$

Write in index notation:

9 $\sqrt{x}$

10 $\sqrt{m}$

11 $\sqrt{36}$

12 $\sqrt[3]{a}$

13 $\sqrt[3]{64}$

14 $\sqrt{x}$

15 $\frac{1}{\sqrt{a}}$

16 $\frac{1}{\sqrt[3]{m}}$

Simplify the following expressions:

17 $(x^4)^{\frac{1}{2}}$

18 $(y^6)^{\frac{1}{3}}$

19 $(y^{\frac{1}{2}})^2$

20 $\sqrt{m^8}$

21 $m^{\frac{1}{2}} \times m^{\frac{1}{2}}$

22 $a^{\frac{3}{2}} \times a^{\frac{1}{2}}$

23 $y^{\frac{1}{3}} \times y^{\frac{2}{3}}$

24 $(a^2b^4)^{\frac{1}{2}}$

25 $(x^2y)^{\frac{1}{2}}$

26 $(9x^4)^{\frac{1}{2}}$

27 $(4x^6)^{\frac{1}{2}}$

28 $(16m^4)^{\frac{1}{4}}$

29 $(8a^6)^{\frac{1}{3}}$

30 $\sqrt[3]{16m^{10}}$

31 $\sqrt[3]{8a^9}$

32 $5a^{\frac{1}{2}} \times 3a^{\frac{1}{2}}$

33 $9k^{\frac{2}{3}} \times 2k^{\frac{1}{3}}$

34 $(4a^2b^6)^{\frac{1}{2}}$

35 $(8x^3y^3)^{\frac{1}{3}}$

36 $2y^3 \times (y^{\frac{1}{2}})^4$

The Fractional Index $\frac{m}{n}$ is made up of 2 powers.

eg $8^{\frac{2}{3}} = (8^{\frac{1}{3}})^2$ or The index $\frac{2}{3} = \frac{1}{3} \times 2 = 2 \times \frac{1}{3}$
 $= (\sqrt[3]{8})^2$ $(8^{\frac{1}{3}})^2 = \sqrt[3]{64}$

C Evaluate, giving your answers as whole numbers or fractions:

1 $9^{\frac{2}{2}}$

2 $16^{\frac{3}{4}}$

3 $27^{\frac{2}{3}}$

4 $4^{\frac{5}{2}}$

5 $9^{-\frac{1}{2}}$

6 $27^{-\frac{1}{3}}$

Write using a fractional index:

7 $\sqrt{m^3}$

8 $\sqrt[3]{m^2}$

9 $\sqrt[3]{a^4}$

10 $\sqrt{a^5}$

11 $\sqrt[4]{x^3}$

12 $\sqrt[5]{x^2}$

Simplify these expressions:

13 $\sqrt[3]{m^6}$

14 $\sqrt{16a^4}$

15 $\sqrt[3]{8x^{12}}$

16 $y^{\frac{2}{3}} \times y^{\frac{1}{3}}$

17 $4y^{\frac{3}{4}} \div 2y^{\frac{1}{2}}$

18 $8m^{\frac{1}{2}} \times 3m^{-\frac{1}{2}}$

19 $(16a^4)^{\frac{3}{2}}$

20 $(4x^2)^{\frac{5}{2}}$

21 $(8y^9)^{\frac{2}{3}}$

22 $(25m^6)^{\frac{3}{2}}$

23 $(16x^8)^{\frac{3}{4}}$

24 $(9y^2)^{-\frac{1}{2}}$

25 $(27a^6)^{-\frac{1}{3}}$

26 $(a^{\frac{1}{2}} \times b^{\frac{2}{3}})^6$

27 $a^{\frac{3}{4}} \times a^{\frac{1}{4}} \times a^{-\frac{3}{2}}$

28 $(16x^6)^{\frac{1}{2}} \times (2x)^{-3}$

29 $\frac{3a \times 9y^{\frac{1}{2}}}{(9y)^{\frac{1}{2}}}$

30 $\sqrt[3]{a^4} \div \sqrt[6]{a^5}$